Specialized Password-Hashing Functions

Goal: Protect passwords against brute-force attacks

Principle: Make each guess slow i.e computationally expensive

- By including a mandatory salt
- By making the hash function CPU and memory intensive
- By having multiple passes that cannot be parallelized using GPUs or specialized hardware

◉ SHA2 & SHA3 are not suitable hash functions because too fast

✓ **Argon2** (recommended), scrypt and bcrypt

Password Strength

Number of Characters	Numbers Only	Lowercase Letters	Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters, Symbols
4	Instantly	Instantly	Instantly	Instantly	Instantly
5	Instantly	Instantly	46 mins	2 hours	3 hours
6	Instantly	37 mins	2 days	5 days	1 week
7	Instantly	16 hours	3 months	10 months	2 years
8	Instantly	2 weeks	12 years	50 years	132 years
9	2 hours	1 year	636 years	3k years	9k years
10	20 hours	32 years	33k years	191k years	645k years
11	1 week	839 years	1m years	11m years	45m years
12	3 months	21k years	89m years	737m years	3bn years
13	2 years	567k years	4bn years	45bn years	221bn years
14	23 years	14m years	241bn years	2tn years	15tn years
15	229 years	383m years	12tn years	175tn years	1qd years
16	2k years	9bn years	653tn years	10qd years	75qd years
17	22k years	259bn years	33qd years	675qd years	5qn years
18	228k years	6tn years	1qn years	41qn years	372qn years

Time it takes a hacker to brute force your password in 2026

Hardware: 16 x RTX 5090
Password hash: bcrypt (10)

Hive Systems

Read more and download at
hivesystems.com/password